

Project Design Phase-II Technology Stack (Architecture & Stack)

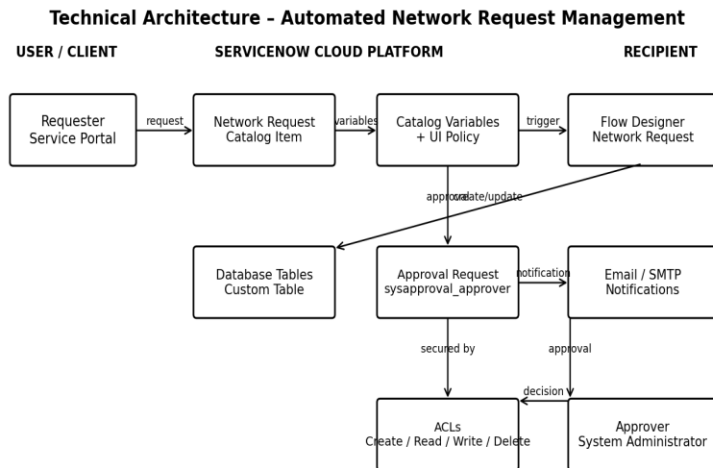
Date	16 September 2026
Team ID	Individual Project
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The technical architecture below represents the implemented ServiceNow solution and the technologies/components used to automate Network Request processing.

Architecture: Automated Network Request Management in ServiceNow

Reference: ServiceNow platform and Flow Designer architecture



Flow path: Submit Request → Get Catalog Variables → Create Database Record → Send Email → Ask for Approval → Approved/Rejected → Update Record

Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interface through which users submit and track Network Requests.	ServiceNow Service Portal / Service Catalog
2.	Application Logic-1	Collects requester and network request information using catalog variables.	ServiceNow Catalog Item + Variable Set
3.	Application Logic-2	Dynamically displays the Existing ID field based on Type of Connection.	ServiceNow Catalog UI Policy
4.	Application Logic-3	Automates variable retrieval, record creation, email notification and approval processing.	ServiceNow Flow Designer / Workflow Studio
5.	Database	Stores submitted network request information in the custom Database Tables table.	ServiceNow custom table (u_database_tables)
6.	Cloud Database	ServiceNow platform provides managed storage for request and approval records.	ServiceNow Platform Database
7.	External API-2	Email delivery uses the ServiceNow email/SMTP infrastructure.	ServiceNow Email / SMTP
8.	Infrastructure (Server / Cloud)	The application is developed, configured and executed on a ServiceNow instance.	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The implemented solution primarily uses ServiceNow's platform configuration and workflow capabilities rather than an external open-source framework.	ServiceNow Platform / Flow Designer
2.	Security	Access to the custom Database Tables is controlled using ServiceNow ACLs. Create, Read, Write and Delete ACLs are configured for the table.	ServiceNow ACLs, Roles and Access Controls
3.	Scalable Architecture	The catalog and Flow Designer approach supports multiple Network Requests and can be extended with additional flows, fields, approvals and tasks.	ServiceNow Service Catalog + Flow Designer
4.	Availability	The application runs on the ServiceNow cloud platform and is accessed through the Service Portal / Service Catalog.	ServiceNow Cloud Platform
5.	Performance	Automation reduces manual processing by retrieving variables, creating records, sending notifications and routing approvals through one flow.	Flow Designer + Service Catalog

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

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<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>